

Maths 141H – Homework 1

Due: **Beginning of class on 02/28/2024**

Rules

- This homework consists of **8 questions** (of unequal difficulty) all carrying equal weight. Please answer all questions. Some parts of questions are **Extra Credit**, and they are worth *an additional* 10% (so it is possible to get up to 110%).
- Work in groups of **2-3 people**. Each group should submit **a single solution**, clearly indicating the group members (names and UIDs are required). **Groups must have been decided by 02/21** and announced to me by email, before the end of the working day. If you cannot find a group, let me know. **All members of the same group will receive the same grade.**
- Submit your answers at the beginning of class on 02/28/2024. Ensure your answers are legible (illegible parts will be ignored), and to help with the grading,¹ please answer each question starting from a fresh page and then staple everything together.
- **Good luck!**

¹A happy grader is a generous grader! Keep this in mind.

Question 1 – Volume of an Obelisk

1. An *obelisk* is a shape consisting of two parts: a *square frustum* F , and a *square pyramid* S on top of F .

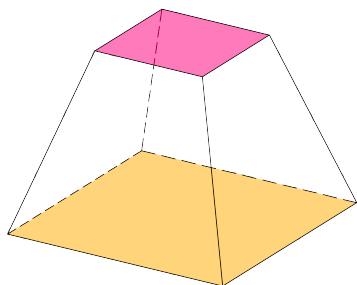


Figure 1: A square frustum.



Figure 2: A square pyramid.

Consider the following obelisk $O = (F, S)$:

- The base of F has side length n_1 meters. The top of F has side length n_2 meters, and the height of F is h_1 meters. The height of S is h_2 meters.

Using the method of cross-sections, write down a formula for the volume V of O . Justify your answers!

2. Look up the formulas for the volumes of a square frustum and a square pyramid. Use these to check your answers in 1.
3. A *round obelisk* is similar to an obelisk, but instead of a square frustum and a square pyramid, it is built of the frustum of a cone F_c and a cylinder C .

Consider the following round obelisk $O = (F_c, C)$:

- The base of F_c has radius r_1 meters. The top of F_c has radius r_2 meters, and the height of F_c is h_1 meters. The height of C is h_2 meters.

Write down a formula for the volume V of O .

4. **Extra Credit:** The Washington Monument has the shape of an obelisk. Its sides slant gradually inward as they rise to the base of the small pyramid at the top. The monument's base is square, 16.80 meters on a side. At the base of the small pyramid, 152.49 meters above ground, the walls are 10.50 meters on a side. Finally, the small pyramid is 16.79 meters tall. Using your answer from the previous part (or otherwise), find the total volume V of the monument.

Question 2 – Revolution 1

1. Suppose f is continuous on $[a, b]$ and the graph of f lies above the line $y = c$. Write down a formula for the volume V of the solid obtained by revolving about the line $y = c$ the region between the graph of f and the line $y = c$ on $[a, b]$.
2. Using part 1 (or otherwise) find the volume V of the solid obtained by revolving about the line $y = -1$ the region between the graph of the equation $y = \sqrt{x+1}$ and the line $y = -1$ on the interval $[0, 1]$.
3. Find the volume V of the solid obtained by revolving about the line $y = 1$ the region between the graph of the equation $y = e^{-2x}$ and the x -axis on the interval $[0, 1]$. (*Hint*: This is not a direct application of 1.)
4. Suppose f and g are continuous on $[a, b]$, and let c be such that $c \leq g(x) \leq f(x)$ for $x \in [a, b]$. Write down a formula for the volume V of the solid obtained by revolving about the line $y = c$ the region between the graphs of f and g on $[a, b]$.
5. Using part 4 (or otherwise) find the volume V of the solid obtained by revolving about the line $y = 1$ the region between the graphs of $y = x^2 - x + 1$ and $y = 2x^2 - 4x + 3$.
6. Suppose $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous functions. Give an example to show that unless $f \geq g$ everywhere, the volume of revolution around the x -axis of the region between the graphs of f and g is no longer given by the formula

$$\int_a^b \pi(f(x)^2 - g(x)^2) dx. \quad (1)$$

To fix this problem, notice that the interval $[a, b]$ can be partitioned into subintervals where $f \geq g$ and intervals where $g \geq f$. Formula (1) can be used in the former intervals while for the latter intervals it is enough to switch the positions of f and g in the formula. Use this observation to argue that the correct formula for the volume is given by

$$\int_a^b \pi |f(x)^2 - g(x)^2| dx. \quad (2)$$

Apply formula (2) to calculate the volume of revolution of the region between the functions $f(x) = \cos(x)$ and $g(x) = \sin(x)$ for $x \in [0, 3\pi/4]$.

Question 3 – Revolution 2

1. Let f be continuous and non-negative on $[a, b]$, and assume that $c \leq a$. Let R be the region between the graph of f and the x -axis on $[a, b]$. Find a formula for the volume V of the solid obtained by revolving R about the line $x = c$.
2. Using part 1 (or otherwise) find the volume V of the solid obtained by revolving about the line $x = -1$ the region between the graph of $y = x^4$ and the x -axis on $[0, 1]$.
3. Let f and g be continuous on $[a, b]$, with $g(x) \leq f(x)$ for $x \in [a, b]$. Let R be the region between the graphs of f and g on $[a, b]$, and assume that $c \leq a$. Find a formula for the volume V of the solid obtained by revolving R about the line $x = c$.
4. Using Riemann sums show that the formula you obtained in part 3 is correct.
5. Using part 3 (or otherwise) find the volume V of the solid obtained by revolving about the line $x = -5$ the region between the graphs of $y = x^2 + 4$ and $y = 2x^2 + x + 2$.
6. Suppose $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous functions. Give an example to show that unless $f \geq g$ everywhere, the volume of revolution around the y -axis of the region between the graphs of f and g is no longer given by the formula

$$\int_a^b 2\pi x(f(x) - g(x)) dx. \quad (3)$$

To fix this problem, notice that the interval $[a, b]$ can be partitioned into subintervals where $f \geq g$ and intervals where $g \geq f$. Formula (3) can be used in the former intervals while for the latter intervals it is enough to switch the positions of f and g in the formula. Use this observation to argue that the correct formula for the volume is given by

$$\int_a^b 2\pi x |f(x) - g(x)| dx. \quad (4)$$

Apply formula (4) to calculate the volume of revolution of the region between the functions $f(x) = \cos(x)$ and $g(x) = \sin(x)$ for $x \in [0, 3\pi/4]$.

Question 4 – Gabriel’s Horn

Let f be continuous and non-negative on $[a, \infty)$. If

$$\lim_{b \rightarrow \infty} \int_a^b \pi [f(x)]^2 dx$$

converges, then we say that the volume of the solid obtained by revolving the graph of f about the x -axis is finite. Otherwise the volume is infinite. Similarly, if

$$\lim_{b \rightarrow \infty} \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

converges, then we say that the surface of the solid obtained by revolving the graph of f about the x -axis is finite. Otherwise the volume is infinite.

For the following functions, determine if the volume of the solid obtained by revolving the graph of f about the x -axis is finite, and if it is, find it.

1. $f(x) = \frac{1}{x}$, for $x \geq 1$.
2. $f(x) = \frac{1}{x^r}$, for r a rational number strictly bigger than 1, and $x \geq 1$.
3. $f(x) = \frac{1}{x^r}$, for r a rational number strictly between 0 and 1, and $x \geq 1$.
4. Show that when $f(x) = \frac{1}{x}$ then the surface area of the solid generated by revolving f about the x -axis, for $x \geq 1$ is infinite.
(*Hint:* Instead of trying to work out the integral, use the General Comparison Property: if $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$.)

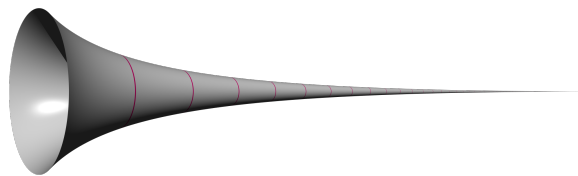


Figure 3: Gabriel’s Horn

5. **Extra Credit:** Prove that it is not possible for a solid of revolution to have finite surface area but infinite volume.

Question 5 – Work work work (work work)

1. A ball weighing 0.2 pound is thrown vertically upward. Its height after t seconds is given by

$$h(t) = 6 + 8t - 16t^2$$

until the ball strikes the ground again.

- (a) Find the work W done on the ball by gravity while the ball descends from its maximum height to the ground.
- (b) Find the work W done on the ball on its descent if it is caught 6 feet above the ground.
2. A thin steel plate in the shape of an isosceles triangle with base 6 feet and height 4 feet is lying flat on the ground.

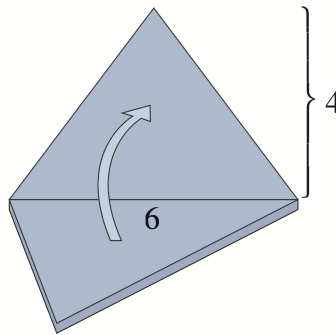


Figure 4: The shape we are trying to lift.

Suppose the weight (in pounds) of any part of the plate equals its area (in square feet). How much work is required in order to raise the plate to a vertical position, assuming the base remains in contact with the ground?

(*Hint:* Let the x axis be perpendicular to the base of the triangle, with the origin on the base. Consider any partition P of $[0, 4]$, and approximate the work by a Riemann sum.)

Question 6 – Tanks and Centres of Gravity

1. (a) An underground tank containing $n\pi$ cubic feet of water has the shape of a right circular cone with its vertex at the bottom. If the radius of the cone is r feet at the top and the height is h feet, find the work W required to pump all the water to a point s feet above the top of the tank.
- (b) Suppose a large gasoline tank has the shape of a half-cylinder n feet in diameter and m feet long. If the tank is full, find the work W necessary to pump all the gasoline out of the tank. Assume the gasoline weighs 42 pounds per cubic foot.
2. Find the centre of gravity of each of the following regions:
 - (a) The region is an isosceles triangle of height h and base b , where the midpoint of the base lies on the origin.
 - (b) The region is a **parabolic sector**:

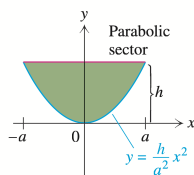


Figure 5: Parabolic Sector

- (c) The region is a **parabolic spandrel**:

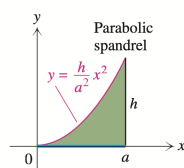


Figure 6: Parabolic Spandrel

- (d) The region is a **finger-like shape** consisting of a rectangle of height h and width w so that the midpoint of its top side lies on the origin together with a semicircle of radius $\frac{w}{2}$ whose centre lies on the origin.

Question 7 – Parameterisations

1. Let $a, b > 0$, and consider the ellipse parameterised by $x = a \cos t$ and $y = b \sin t$ for $0 \leq t \leq 2\pi$. Find a representation of the ellipse in rectangular coordinates.
2. Consider the Folium of Descartes, parameterised by

$$x = \frac{3t}{1+t^3} \text{ and } y = \frac{3t^2}{1+t^3} \text{ for all } t \neq -1$$

Find an equation in rectangular coordinates for the folium.

3. A Lissajous figure is a curve that has a parameterisation of the form:

$$x = \cos(mt) \text{ and } y = \sin(nt) \text{ for } t \in [0, 2\pi]$$

where $m, n \in \mathbb{N}$ are non-zero. Let L_{mn} denote the Lissajous figure for a given pair m, n .

- (a) Suppose that $m = n$. Prove that L_{mn} is a circle.
- (b) Suppose that m and n are both even and n divides m . Prove that the origin is not on L_{mn} .
- (c) Let $m = 4$ and $n = 6$. Prove that the origin is not on L_{mn} .
- (d) Let $m = 3$ and $n = 6$. Prove that the origin is on L_{mn} .

Question 8 – Lengths

1. Let $a, b > 0$, and consider the ellipse parameterised by

$$x = a \cos(t) \text{ and } y = b \sin(t) \text{ for } t \in [0, 2\pi]$$

- (a) Find an integral that represents the circumference C_{ab} of the ellipse.
 - (b) Use the formula obtained in (a) to find C_{ab} when $a = b$.
 - (c) Use the result of (a) to determine $\lim_{b \rightarrow 0^+} C_{ab}$.
2. Consider the parameterised curve given by $x = 1 + \cos(t)$ and $y = 2 - \sin(t)$ for $-2\pi \leq t \leq 2\pi$.
- (a) Sketch this curve in the xy -plane. Include an arrow specifying which direction the curve is traced as the parameter varies from $t = -2\pi$ to $t = 2\pi$.
 - (b) How many times does this curve get traced as t varies from -2π to 2π ?
 - (c) Write down an integral that gives the length of this curve, and then find the length.
3. Let $r > 0$. The equations $x = r \cos^3 t$ and $y = r \sin^3 t$ parameterise an astroid. Draw the graph, and find the length L of the astroid.